

Supplementary File 8. Excluded full texts with reasons

Review Title

Reframing the Malaria-Sickle Cell Disease Relationship in Sub-Saharan Africa as a Bounded Syndemic: A Scoping Review and Evidence Gap Map Purpose

This supplementary file documents sources excluded after full-text assessment and provides the primary reason for exclusion. The purpose is to make the full-text screening process transparent, reproducible, and auditable.

Each source is listed once with one primary exclusion reason. Where more than one reason could apply, the dominant reason is recorded.

Full-Text Exclusion Categories

Code	Exclusion reason	Operational definition
E1	Wrong geography	Source was not conducted in sub-Saharan Africa and did not provide foundational, conceptual, comparative, intervention, or theoretical evidence directly relevant to the review.
E2	Wrong population	Population did not relate to malaria, SCD, HbS, red blood cell polymorphisms, severe anaemia, newborn screening, routine visibility, SCD care, or relevant health-system actors.
E3	Wrong concept	Source did not address any review concept, including

		biological interaction, spatial co-risk, clinical convergence, service-mediated visibility, structural vulnerability, programme response, or syndemic theory.
E4	No malaria relevance	Source addressed SCD or another condition but did not relate to malaria, malaria-endemic settings, malaria-like illness, severe anaemia, malaria-SCD interpretation, or malaria ecology.
E5	No SCD, HbS, or red cell relevance	Source addressed malaria but did not address SCD, SCA, HbSS, HbAS, HbS, sickle cell trait, malaria-relevant red blood cell polymorphisms, or SCD care.
E6	No syndemic relevance	Source was broadly related to malaria, SCD, or health systems but did not contribute to any of the seven evidence domains.
E7	Laboratory-only without review relevance	Source was laboratory-based and did not inform human malaria-SCD biology, red blood cell polymorphisms, clinical interpretation,

		epidemiology, or programme relevance.
E8	Animal-only without human relevance	Source used animal models only and did not provide evidence directly relevant to human malaria-SCD relationships.
E9	Insufficient information for charting	Full text did not provide enough information on population, context, methods, results, or relevance to enable data charting.
E10	Duplicate or overlapping report	Source reported the same dataset, trial, programme, cohort, or evidence already represented by a fuller or more relevant included source.
E11	Commentary without substantive contribution	Source was an editorial, opinion piece, letter, news item, or commentary without substantive empirical, conceptual, theoretical, policy, or programme contribution.
E12	Full text unavailable	Full text could not be retrieved after reasonable attempts, and the abstract did not provide enough information for inclusion.

E13	Wrong source type	Source type was outside eligibility, for example conference title only, abstract without results, non-substantive webpage, or promotional material.
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Table 1. Excluded Full Texts With Primary Reason for Exclusion

Exclusion ID	Citation	Country or region	Source type	Reason code	Primary reason for exclusion	Reviewer notes
EX001	Dondorp et al., 2010, Artesunate versus quinine in the treatment of severe falciparum malaria in African children, AQUAMAT	Multicountry Africa	Randomised trial	E5	No SCD, HbS, or red cell relevance	Important malaria treatment trial, but did not assess SCD, HbS, sickle trait, red cell polymorphisms, or SCD service visibility.
EX002	RTS,S Clinical Trials Partnership	Seven African countries	Vaccine trial	E5	No SCD, HbS, or red cell relevance	Malaria vaccine trial without

	, 2011, First results of phase 3 trial of RTS,S/AS 01 malaria vaccine in African children					SCD, HbS, red cell polymorphi sm, or malaria- SCD convergenc e analysis.
EX003	RTS,S Clinical Trials Partnership , 2012, Phase 3 trial of RTS,S/AS 01 malaria vaccine in African infants	Seven African countries	Vaccine trial	E5	No SCD, HbS, or red cell relevance	Malaria vaccine evidence only, with no SCD or HbS- related subgroup analysis relevant to the review.
EX004	RTS,S Clinical Trials Partnership , 2015, Efficacy and safety of	Multicount ry Africa	Vaccine trial	E5	No SCD, HbS, or red cell relevance	Retained at screening because of malaria and African child health

	RTS,S/AS 01 malaria vaccine with or without booster dose					relevance, but excluded after full text because it did not address SCD, HbS, or red cell traits.
EX005	Maitland et al., 2011, Mortality after fluid bolus in African children with severe infection, FEAST	Uganda, Kenya, Tanzania	Randomise d trial	E5	No SCD, HbS, or red cell relevance	Severe paediatric illness trial in malaria- endemic settings, but did not contribute to malaria- SCD clinical convergen ce, SCD under- recognition , or red cell polymorphi sm evidence.

EX006	Pryce et al., 2018, Insecticide-treated nets for preventing malaria	Global malaria-endemic settings, many African trials	Cochrane review	E5	No SCD, HbS, or red cell relevance	Malaria prevention review without SCD, HbS, red cell polymorphism, or SCD care relevance.
EX007	Sinclair et al., 2009, Artemisinin-based combination therapy for treating uncomplicated malaria	Global malaria-endemic settings	Systematic review	E5	No SCD, HbS, or red cell relevance	Malaria treatment evidence without SCD or red cell polymorphism relevance.
EX008	Abba et al., 2011, Rapid diagnostic tests for diagnosing uncomplicated <i>P. falciparum</i> malaria in	Malaria-endemic countries	Diagnostic test review	E5	No SCD, HbS, or red cell relevance	Diagnostic evidence for malaria RDTs, but no SCD, HbS, severe anaemia attribution, or malaria-

	endemic countries					SCD interpretation.
EX009	Okebe et al., 2014, Intermittent preventive treatment for malaria in infants	Malaria-endemic settings	Systematic review	E5	No SCD, HbS, or red cell relevance	Malaria prevention review without SCD or red cell trait relevance.
EX010	Meremikwu et al., 2012, Intermittent preventive treatment for malaria in children living in areas with seasonal transmission	Seasonal malaria transmission settings	Systematic review	E5	No SCD, HbS, or red cell relevance	Malaria prevention evidence did not address SCD, HbS, red cell polymorphisms, or SCD service planning.
EX011	Olotu et al., 2016, Seven-year efficacy of RTS,S/AS	Kenya, Tanzania	Vaccine follow-up study	E5	No SCD, HbS, or red cell relevance	Malaria vaccine follow-up without SCD or red

	01 malaria vaccine					cell polymorphism analysis.
EX012	Bousema et al., 2014, Hitting hotspots: spatial targeting of malaria for control and elimination	Global malaria relevance	Review	E6	No syndemic relevance	Spatial malaria control review, but did not address HbS, SCD, red cell polymorphisms, severe anaemia, or SCD programme planning.
EX013	Tusting et al., 2015, Housing improvements and malaria risk in sub-Saharan Africa	Sub-Saharan Africa	Systematic review	E5	No SCD, HbS, or red cell relevance	Malaria exposure and housing study excluded because it did not connect to HbS, SCD, or SCD care.

EX014	Smith et al., 2007, Standard antimalarial drugs and malaria treatment policy	Global malaria relevance	Review	E5	No SCD, HbS, or red cell relevance	General malaria treatment policy without review-domain contribution.
EX015	Menendez et al., 2010, Malaria in pregnancy	Malaria-endemic settings, Africa relevance	Review	E5	No SCD, HbS, or red cell relevance	Pregnancy malaria review without SCD, HbS, or sickle-related maternal-child pathway evidence.
EX016	Charache et al., 1995, Effect of hydroxyurea on painful crises in sickle cell anaemia	United States	Randomised trial	E1	Wrong geography	Foundational hydroxyurea trial, but excluded from final D1-D7 inventory because SSA-

						specific hydroxyurea evidence was available and this trial did not address malaria-endemic care systems.
EX017	Wang et al., 2011, Hydroxy carbamide in very young children with sickle-cell anaemia, BABY HUG	United States	Randomised trial	E1	Wrong geography	Important paediatric hydroxyurea trial, but excluded because it did not address SSA, malaria exposure, or African service delivery.
EX018	DeBaun et al., 2014, Controlled trial of	United States, Canada, United	Randomised trial	E1	Wrong geography	SCD transfusion trial outside

	transfusions for silent cerebral infarcts in sickle cell anaemia	Kingdom, France				SSA, without malaria-endemic context or screening-to-care relevance for this review.
EX019	Adams and Brambilla, 2005, Discontinuing prophylactic transfusions used to prevent stroke in SCD	United States	Randomised trial	E1	Wrong geography	Stroke prevention transfusion trial excluded because it did not address SSA, malaria-SCD convergence, or service visibility.
EX020	Ataga et al., 2017, Crizanlizumab for the prevention	United States, Brazil, Jamaica	Randomised trial	E1	Wrong geography	SCD drug trial outside SSA and not linked

	of pain crises in SCD					to malaria-endemic service systems or screening-to-care planning.
EX021	Vichinsky et al., 2019, A phase 3 randomized trial of voxelotor in SCD	Multinational, not SSA-focused	Randomised trial	E1	Wrong geography	Drug trial excluded because it did not address SSA, malaria, SCD visibility, or African programme response.
EX022	Walters et al., 1996, Bone marrow transplantation for sickle cell disease	United States	Clinical intervention study	E1	Wrong geography	Curative SCD intervention outside SSA, without malaria-endemic context or programme relevance

						to review question.
EX023	Bhatia and Walters, 2008, Hematopoietic cell transplantation for thalassemia and sickle cell disease	Global, high-resource care focus	Review	E1	Wrong geography	Transplantation-focused review did not contribute to SSA malaria-SCD syndemic planning framework.
EX024	Rees et al., 2010, Sickle-cell disease	Global	Review	E4	No malaria relevance	General SCD review excluded because more directly relevant Africa-specific and malaria-linked SCD sources

						were included.
EX025	Bunn, 1997, Pathogenes is and treatment of sickle cell disease	Global	Review	E4	No malaria relevance	General SCD pathophysiology review without SSA, malaria, visibility, or programme response relevance.
EX026	Platt et al., 1994, Pain in sickle cell disease: rates and risk factors	United States	Cohort study	E1	Wrong geography	SCD pain cohort outside SSA without malaria or African service-system relevance.
EX027	Ballard et al., 2017, Long-term impact of	United States	Cohort or follow-up study	E1	Wrong geography	Hydroxyurea outcome study outside SSA, not

	hydroxyurea in SCD					linked to malaria-endemic care systems.
EX028	Kwiatkowski, 2005, How malaria has affected the human genome and what human genetics can teach us about malaria	Global	Review	E10	Duplicate or overlapping report	Broad malaria-human genetics review. Excluded because the review retained more specific included sources on HbAS, HbS mapping, HbC, G6PD, ABO, Dantu, and MalariaGEN.
EX029	Cserti and Dzik, 2007, The	Global	Review	E10	Duplicate or	Excluded as overlapping

	ABO blood group system and <i>Plasmodium falciparum</i> malaria				overlapping report	g review evidence because primary ABO studies by Rowe et al. and Fry et al. were retained.
EX030	Luzzatto, 2012, G6PD deficiency and malaria selection	Global	Review	E10	Duplicate or overlapping report	Excluded as overlapping conceptual review because primary and meta-analytic G6PD evidence was retained.
EX031	Haldane, 1949, Disease and evolution	Global theoretical relevance	Historical theoretical paper	E10	Duplicate or overlapping report	Foundational but not charted because the review retained

						more directly relevant HbS malaria-selection and mapping evidence.
EX032	Lell et al., 1999, Severe malaria in Gabonese children: clinical and laboratory features	Gabon	Hospital-based malaria study	E5	No SCD, HbS, or red cell relevance	Severe malaria clinical study without SCD, HbS, red cell polymorphism, or under-recognition analysis.
EX033	Dzeing-Ella et al., 2005, Severe falciparum malaria in Gabonese children	Gabon	Hospital-based clinical study	E5	No SCD, HbS, or red cell relevance	Severe malaria phenotype study excluded because it did not address SCD, HbS,

						or malaria-SCD attribution.
EX034	Newton et al., 1997, Severe anaemia in children living in a malaria-endemic area	Kenya	Hospital-based clinical study	E5	No SCD, HbS, or red cell relevance	Severe anaemia study retained at screening but excluded because SCD or HbS recognition was not part of the analysis.
EX035	Calis et al., 2008, Severe anaemia in Malawian children	Malawi	Case-control study	E5	No SCD, HbS, or red cell relevance	Important severe anaemia aetiology study, but excluded because it did not contribute malaria-SCD or HbS-

						specific evidence.
EX036	Phiri et al., 2008, Severe anaemia and cerebral malaria in African children	Malawi	Clinical study	E5	No SCD, HbS, or red cell relevance	Clinical malaria and anaemia study without SCD, HbS, or red cell polymorphism analysis.
EX037	Berkley et al., 2005, Bacteraemia among children admitted to a rural hospital in Kenya	Kenya	Hospital-based cohort	E3	Wrong concept	Important paediatric infection study, but did not address malaria-SCD convergence, HbS, SCD visibility, or programme response.
EX038	Scott et al., 2011,	Kenya	Time-series or	E5	No SCD, HbS, or red	Malaria-infection

	Relation between falciparum malaria and bacteraemia in Kenyan children		hospital-based analysis		cell relevance	interaction study excluded because it did not address SCD, HbS, or sickle-related vulnerability.
EX039	Roca-Feltrer et al., 2008, Estimating malaria burden in Africa using routine data	Africa	Routine-data malaria study	E10	Duplicate or overlapping report	Excluded because more directly relevant routine malaria spatial methods were represented by Dalrymple et al. and Alegana et al.
EX040	Malaria Indicator Survey	SSA country-specific	Survey report	E5	No SCD, HbS, or red	Malaria indicator survey

	country report, example national malaria- only report				cell relevance	reports were excluded when they provided malaria exposure data only and did not address HbS, SCD, severe anaemia attribution, or SCD service planning.
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Table 2. Summary of Full-Text Exclusions by Reason

Reason code	Exclusion reason	Number of full texts excluded
E1	Wrong geography	11
E2	Wrong population	0
E3	Wrong concept	1
E4	No malaria relevance	2
E5	No SCD, HbS, or red cell relevance	19
E6	No syndemic relevance	1
E7	Laboratory-only without review relevance	0

E8	Animal-only without human relevance	0
E9	Insufficient information for charting	0
E10	Duplicate or overlapping report	6
E11	Commentary without substantive contribution	0
E12	Full text unavailable	0
E13	Wrong source type	0
Total	All excluded full texts	40